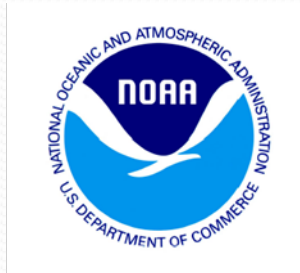


Overview of the Atlantic Sea Scallop Resource in the Gulf of Maine

Research Track Peer Review
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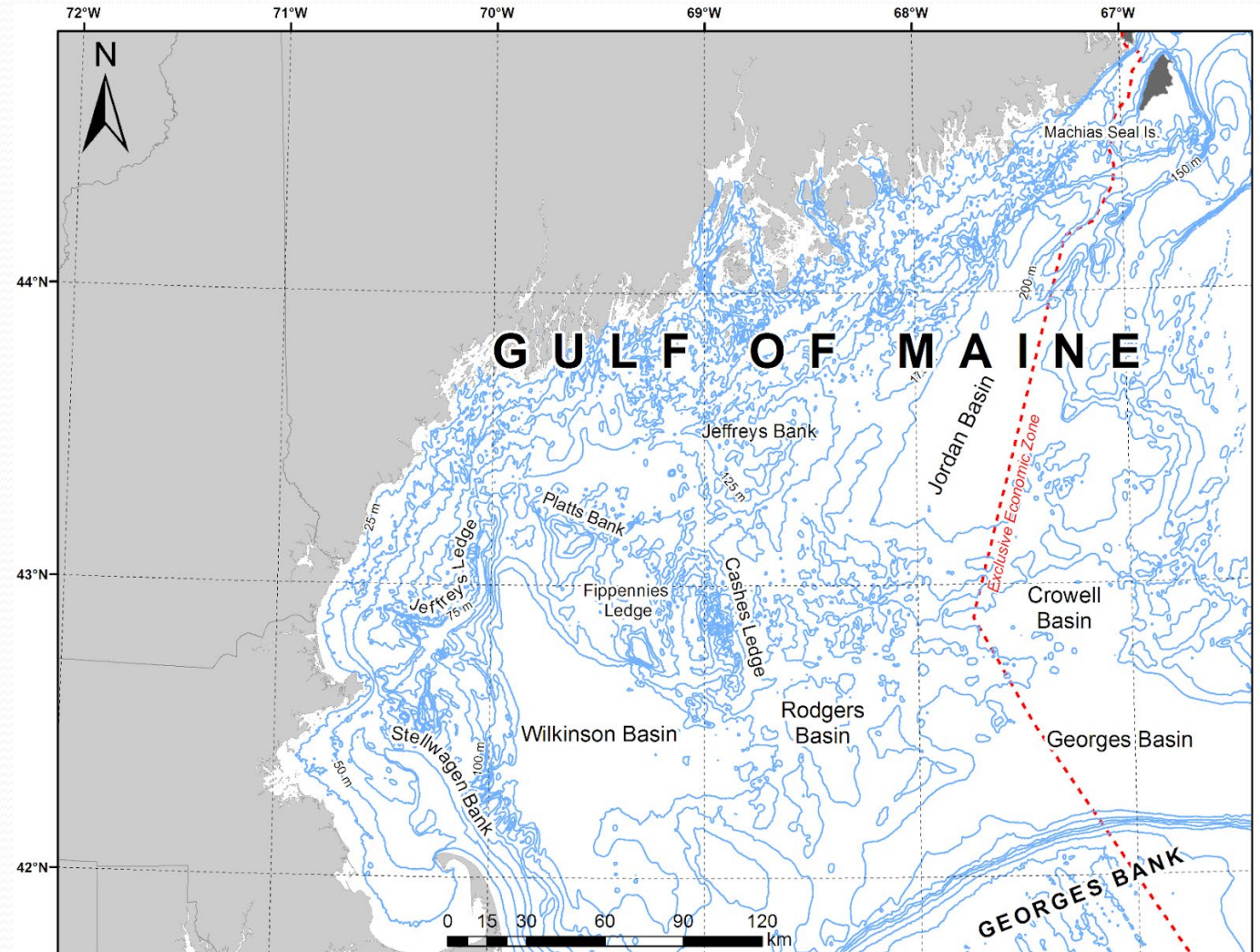


Working Paper (Appendix) Overview

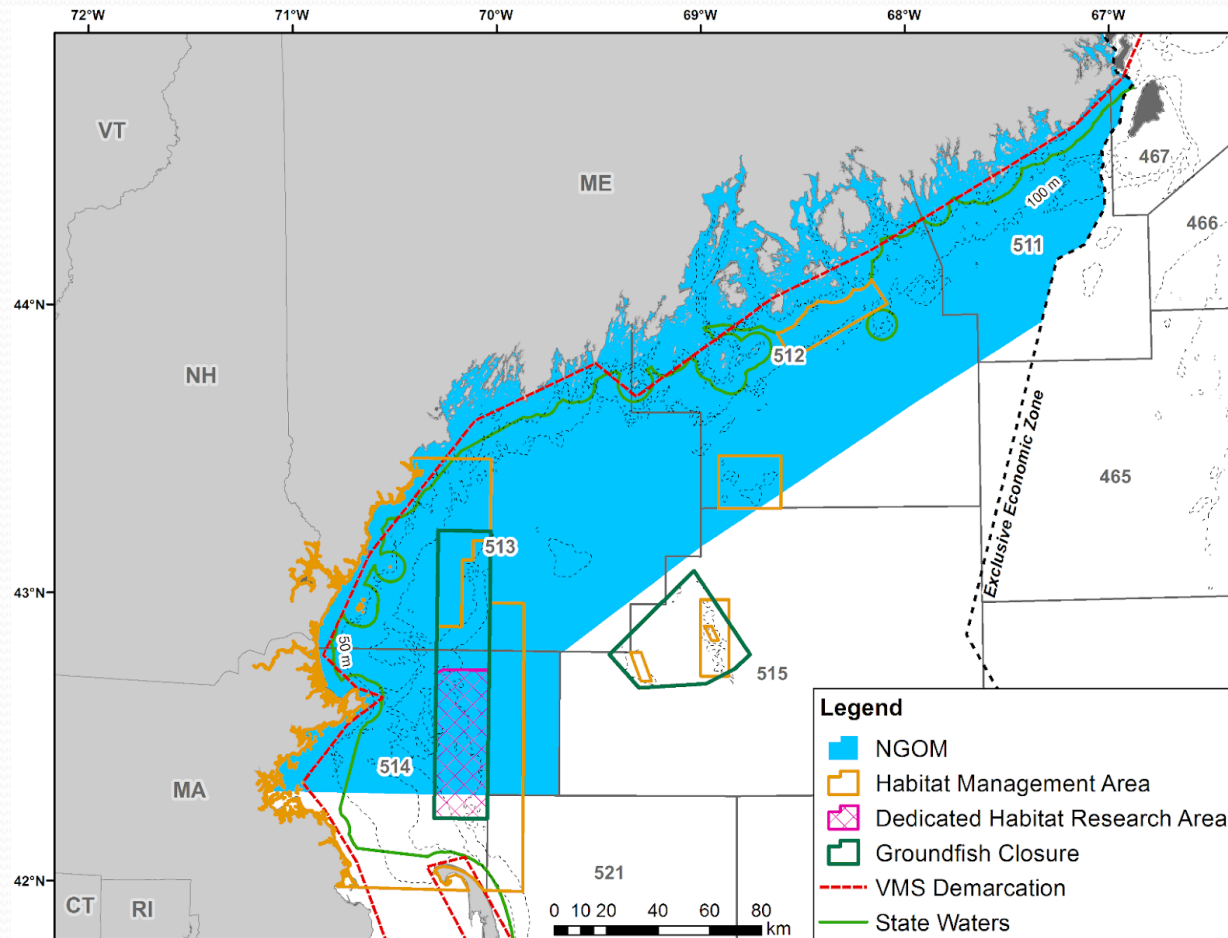
Sections	Areas of Focus
Gulf of Maine Region	Overview
Fishery Independent Surveys	SMAST drop camera optical survey Maine DMR dredge survey (state / federal) Maine/New Hampshire trawl survey
Catch advice – Methods and Procedures	Recent approaches and recommended procedures
Biological Reference Points (BPRs)	Data needs to support the develop of BPRs
Federal Fishery Management	Northern Gulf of Maine Management Unit
Fishery Dependent Data	Seasons, Landings, Observer Data, Vessel Monitoring System

Scallops & Gulf of Maine

- Characterized by banks, ledges, and deep basins.
- Limited sand, pebble, cobble habitats (generally banks & ledges).
- Scallops can occur in both inshore shallower areas compared to Georges Bank and the Mid-Atlantic.
- Inshore state waters fisheries, and offshore federal fishery.
- Boom and bust cycles.
- Not considered in CASA or SYM,
 - GB Proxy in Catch Setting.



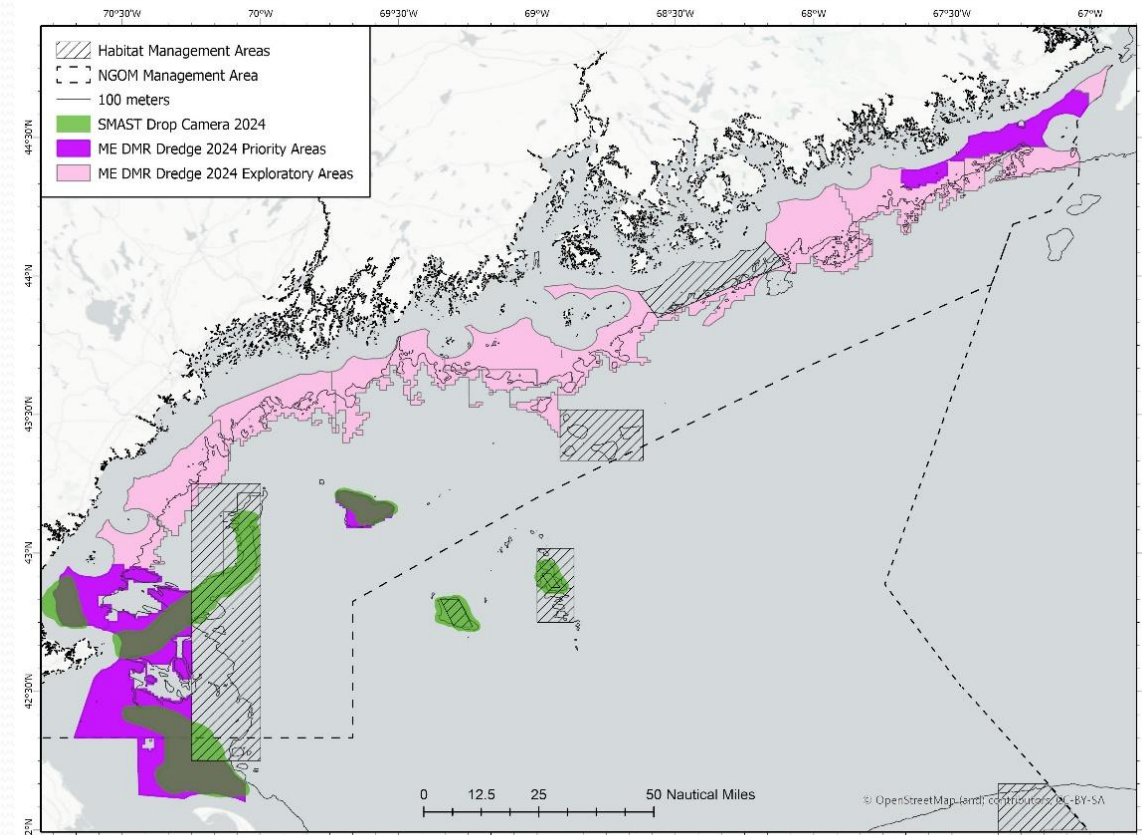
Scallop Management Boundaries



- In the Gulf of Maine, commercial fishing is excluded from habitat management area, groundfish closures, and dedicated habitat research areas.
- The NGOM Management Area covers the majority of inshore areas, starting at 42° 20' N (roughly Boston, MA) and extending north to the Canadian border.

Fishery Independent Surveys

- Periodic surveys in the 1980's.
- No formal survey strata → Areas of Interest.
- Growing time series of scallop survey data that has largely been supported through a research set-aside. Supports development of catch limits.
 - SMAST Drop Camera
 - Maine DMR Dredge
- Other data sources include state waters trawl surveys and federal trawl survey.



Map 2 - Survey Domains for the entire Gulf of Maine from 2024 SMAST Drop Camera and Maine DMR Dredge Surveys.

SMAST Surveys

- Drop camera survey with a fine-scale systematic grid (1 km).
- Methodology described in TOR 3.
 - Suited for GOM → Fixed gear
- Biological sampling starting in 2022.
- Estimates of abundance, collecting meat weight and gonad weights.
- 12 years of surveys between 2009 and 2023. Variation in survey coverage.
- Anticipate continued surveys in the region.

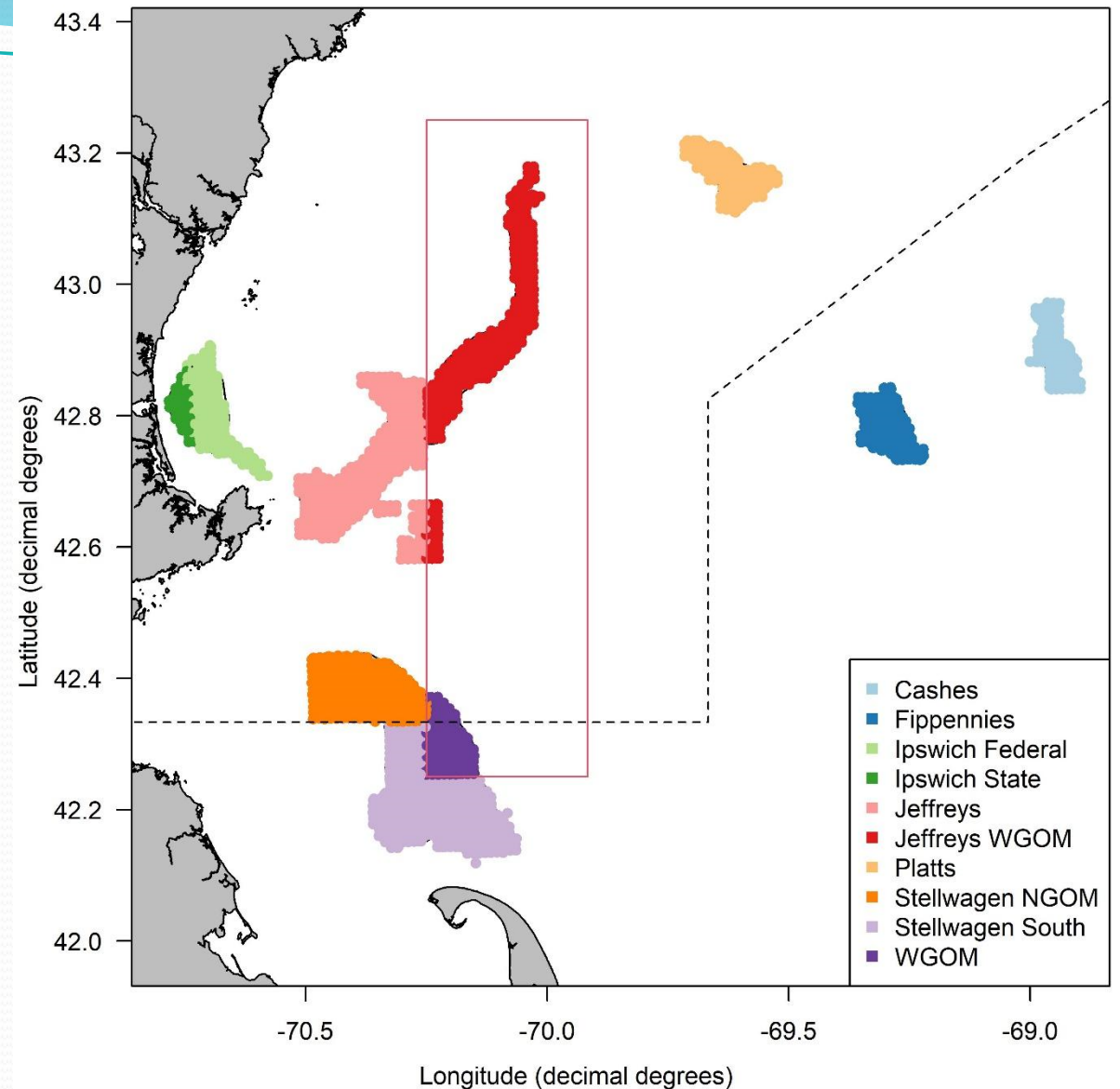


Figure 5. The areas and names of the Gulf of Maine where SMAST estimated scallop abundance and biomass. The red box surrounds the Western Gulf of Maine closure area and the dashed black line denotes the southern and eastern boundaries of the Northern Gulf of Maine management area.

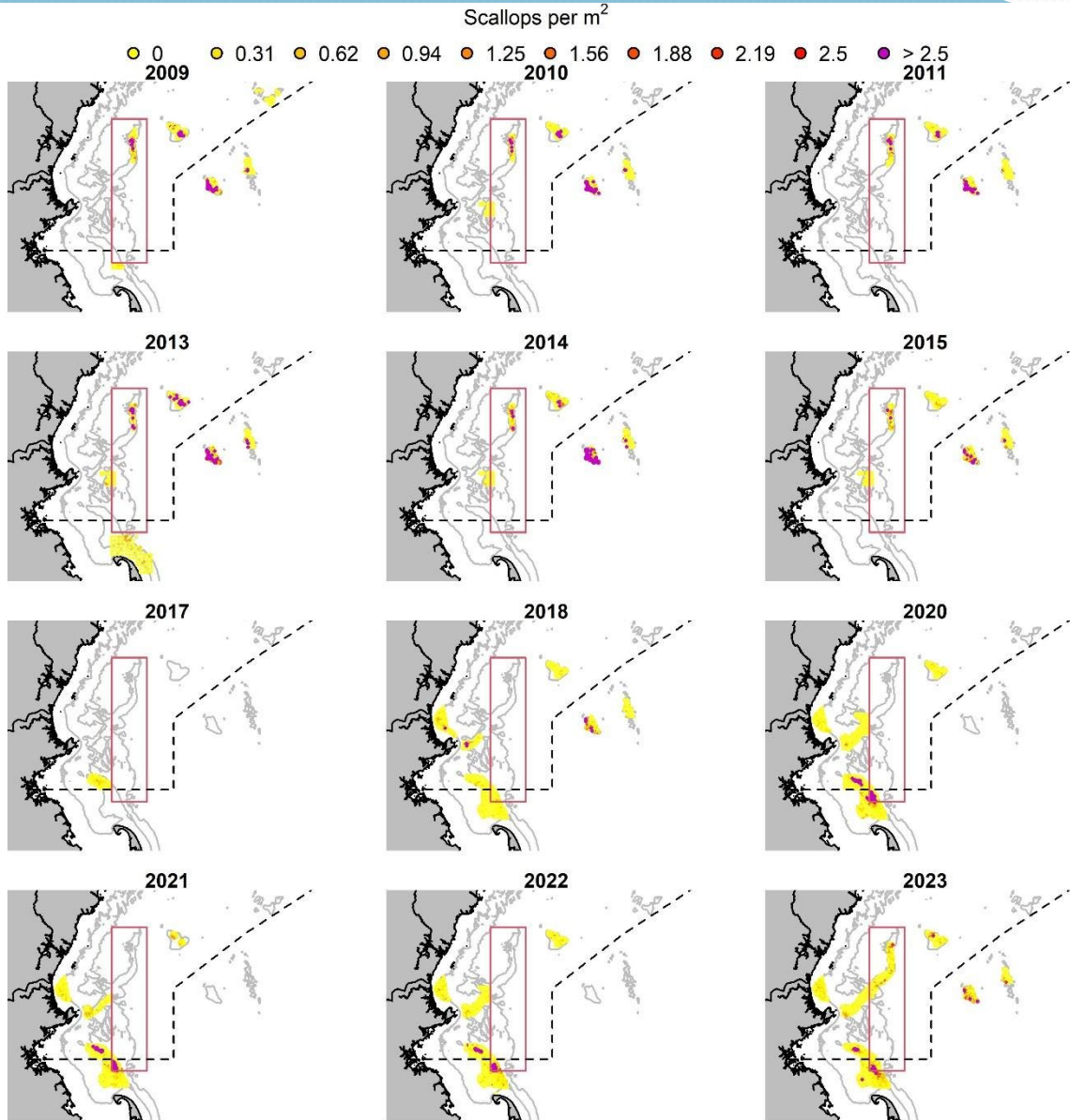


Figure 6. Scallop density (count per m²) heatmap from the SMAST drop camera survey in the Gulf of Maine by year. All sizes of scallops measured are represented.

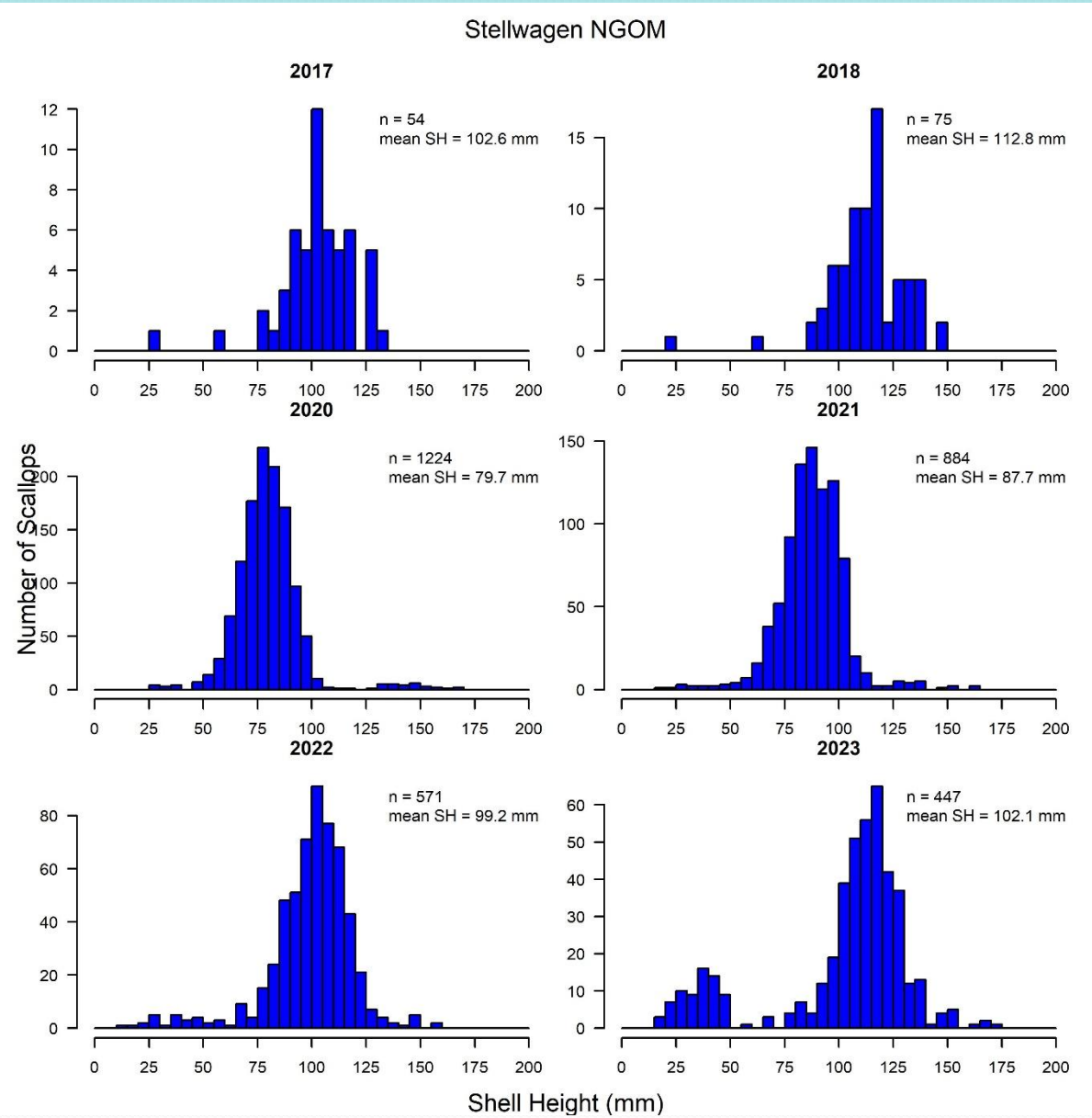


Figure 17. Scallop shell height frequencies from the SMAST drop camera survey on the part of Stellwagen Bank that is within the Northern Gulf of Maine management area, but outside the Western Gulf of Maine closure area.

Maine DMR

- Scallop dredge surveys in the federal waters of the Gulf of Maine since 2009.
- Survey dredge is a 7-ft wide New Bedford-style chain sweep dredge with 2-inch rings, 1¾-inch head bale, 5-inch twine top, and a 10-inch pressure plate. Rock chains, and it is unlined. 5-minute tows, 3.5 kts.
 - Designed to optimize sampling in the rocky substrates that are commonly encountered in the Gulf of Maine.
- Density-stratified random grid survey.
 - Details in working paper.



Maine DMR - Dredge

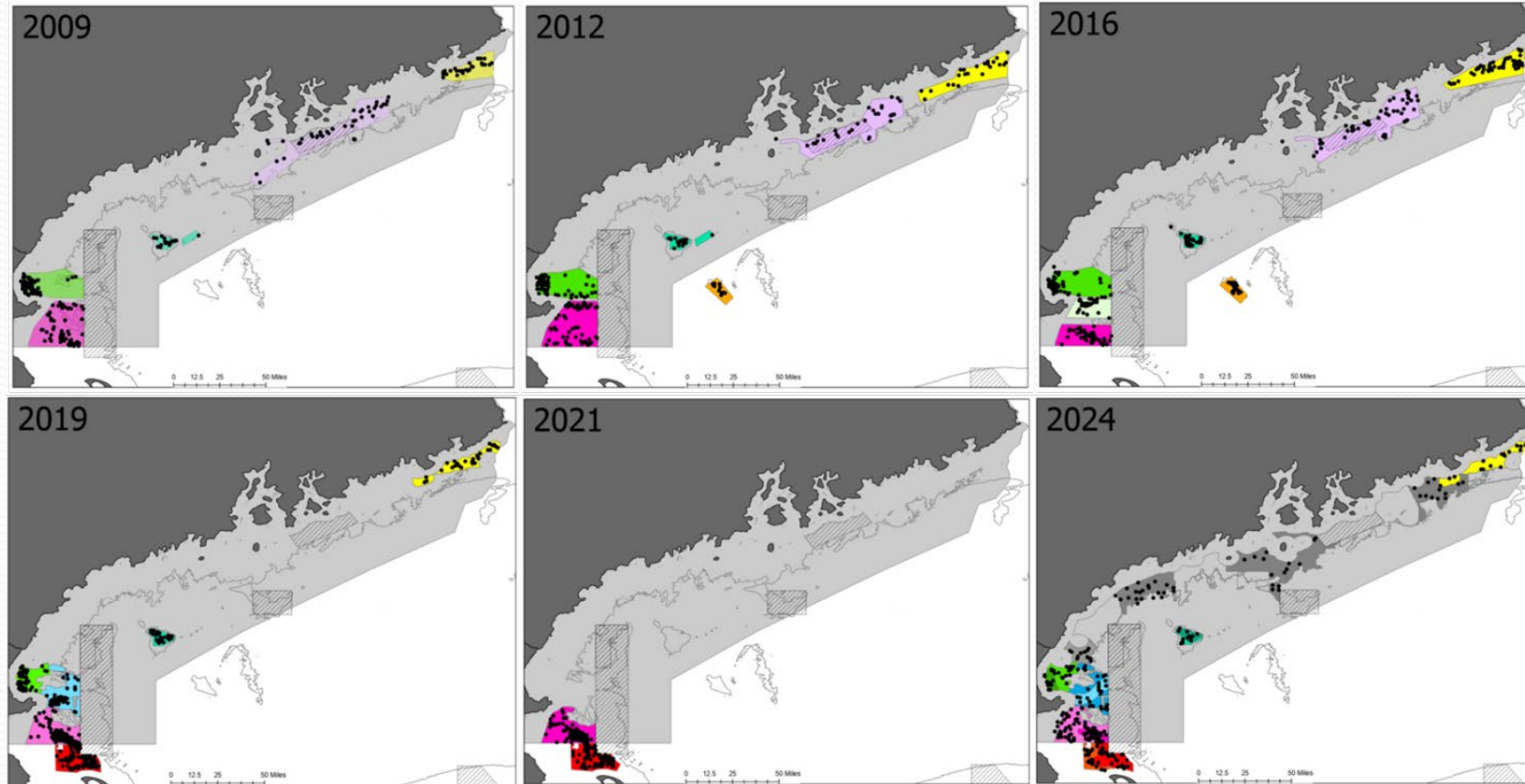


Figure 23. Maine DMR scallop dredge survey domain. The colored areas denote the survey subareas for each given year and the black points are tow locations. The NGOM area is shaded light grey. Yellow=Machias Seal Island, Purple=Mount Dessert Island, Teal=Platts Bank, Orange=Fippennies Ledge, Green=Ipswich Bay, Blue=Jefferys Ledge, Pink=Northern Stellwagen Bank, Red=Southern Stellwagen Bank, Dark Grey=Exploratory Areas.

Shell-Height to Meat-Weight Equations

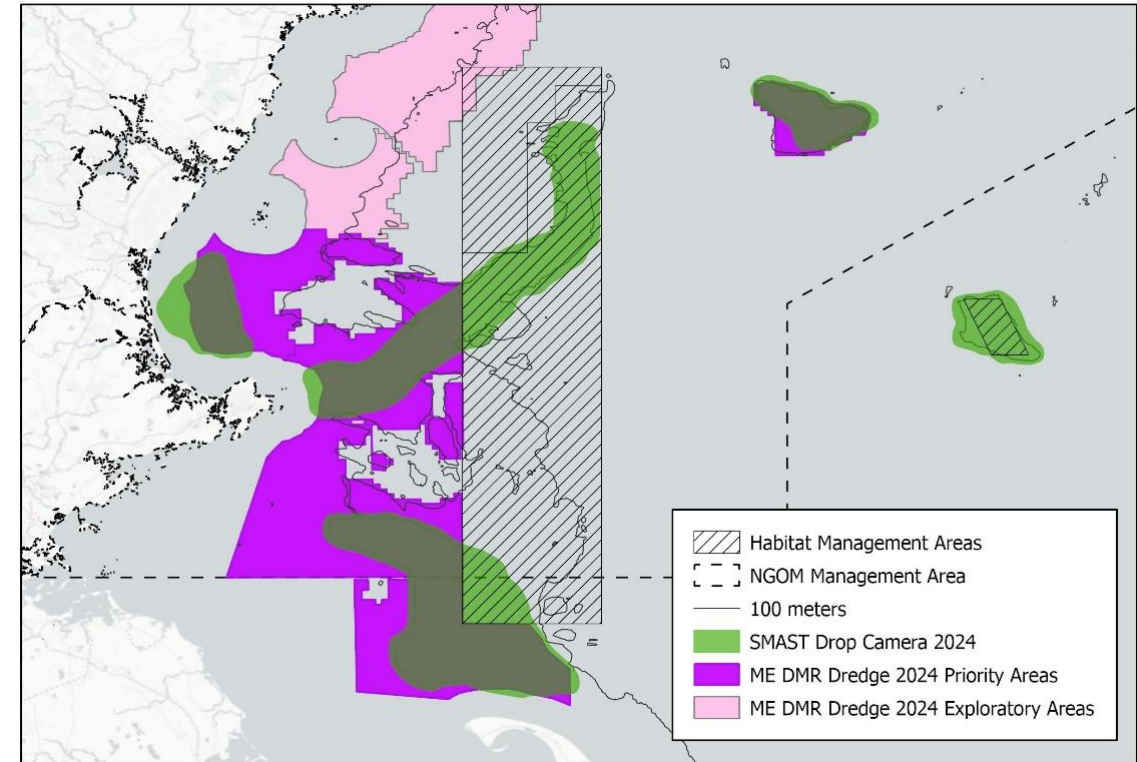
- SH-MW equations used to estimate biomass in the area have evolved over time as additional data has become available.

Years	Name	Equation
2009 to 2017	SAW 50	$W = \exp(9.6771 + 2.8387 \cdot \ln(\text{SH}) - 0.5084 \cdot \ln(\text{depth}) - 4.7629 \cdot \ln(\text{latitude}))$
2018 to 2020	SARC/SAW 65 (GB Open)	$W = \exp(-6.69 + 2.878 \cdot \ln(\text{SH}) - 0.0073 \cdot \text{depth} - 0.073 \cdot \text{latitude})$
2021 to 2022	Hart 2020 (all other areas)	$W = \exp(-281.91 + 72.42 \cdot \ln(\text{latitude}) + -0.212 \cdot \ln(\text{depth}) + (71.13 - 18.16 \cdot \ln(\text{latitude})) \cdot \ln(\text{SH}))$
2021 to 2022	UMaine/DMR 2021 (Stellwagen areas)	$W = \exp(90.21118 + 2.894163 \cdot \ln(\text{SH}) - 26.79 \cdot \ln(\text{latitude}) - 0.07074 \cdot \ln(\text{depth}))$
2023	SMAST/DMR 2016-2023	$W = \exp(-10.36 + 3.032 \cdot \ln(\text{SH}) - 0.2155 \cdot \ln(\text{depth}) + 0.001 \cdot \text{latitude})$
2024 (Stellwagen, Jeffreys, Ipswich Bay)	SMAST/DMR 2016-2024	$W = \exp(-10.09 + 3.0 \cdot \ln(\text{SH}) - 0.239 \cdot \ln(\text{depth}) + 0.00156 \cdot \text{latitude} + \text{area});$ Stellwagen= 0.082; Ipswich=-0.095; Jeffreys = 0.007
2024 (MSI & Platts Bank)	DMR 2016-2024	$W = \exp(1.284 + 2.94 \cdot \ln(\text{SH}) - 0.2098 \cdot -0.2655 \cdot \text{latitude} + \text{area});$ MSI= 0.0668; Platts Bank=-0.026



Procedures for Developing Catch Advice

- Surveys overlap, but not perfectly. Recent procedure:
 - a) Combine data from areas of survey overlap. The general practice is to average the estimates of abundance and biomass.
 - b) Consider data that are within that same spatial area (e.g. the same bank or ledge) but do not overlap with other independent surveys.
 - c) Since 2018, forward projection method (SAMS)
- Recommend formalizing the short-term procedure for developing catch advice.
 - Utilize fishery independent and dependent data to identify areas of interest.
 - Areas of interest used to standardize survey coverage, use in development of catch advice.
 - Develop more formal survey strata over the long-term.
 - Heterogenous distribution, substantial fixed gear → multiple survey designs and methods.



Map 1 - Survey domains for inshore Gulf of Maine east of Massachusetts and New Hampshire from 2024 SMAST Drop Camera and Maine DMR Dredge Surveys.

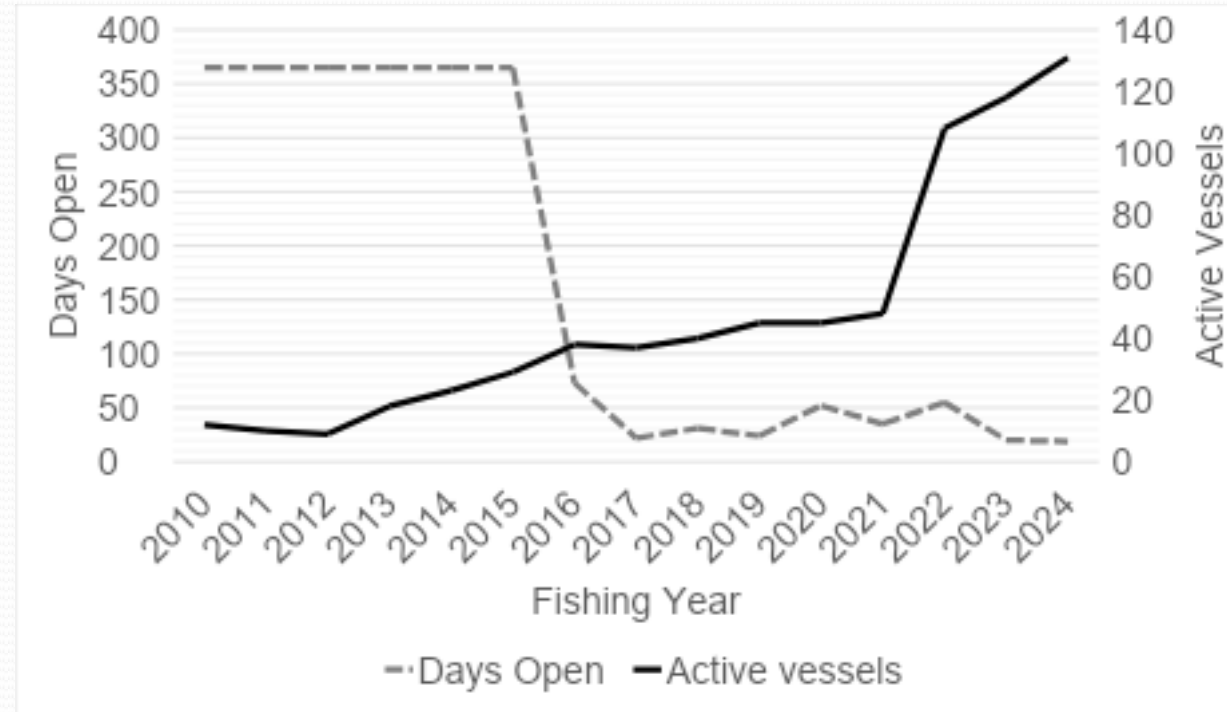
Biological Reference Points



- Gulf of Maine is data poor relative to Georges Bank and the Mid-Atlantic.
- Not considered in CASA or SYM.
- As noted in SARC 65, developing models for the are would require estimates of:
 - Abundance
 - Biomass
 - Growth
 - Natural mortality
 - Indidental and discard mortality
 - Landings
 - Additional life-history information to estimate YPR.
- Continued and expanded data collection since the last research track/benchmark assessment.

Federal Fishery Management in NGOM & Fishery Dependent Data

- Northern Gulf of Maine management area / permit class established in 2008 as small boat fishery, first TACs based on historic catch.
 - Vessels may land 200 pounds of meats per trip.
- 2016 → Growing directed fishery in response to recruitment events on Stellwagen Bank. More management measures, and prioritization of data collection through surveys.
- Key changes implemented in 2022 through Amendment 21 to the Fishery Management Plan.
 - At-sea observers.
 - Harvest sharing arrangement, including set-asides for research, monitoring (observers), and directed fishing.
 - Target F rates must range from 0.15 – 0.25.
- Recent years: Stellwagen Bank. Short seasons and increased participation.



Gulf of Maine – OFL and ABC

- Estimates of biomass and abundance from surveys in the region are included in fishery-wide estimates of OFL and ABC since the implementation of Amendment 21 (starting in 2022).
- No reference points for this region, GB reference points used as proxy.
- Georges Bank F_{MSY} estimates from the 2020 management track assessment ($F=0.46$ for OFL, $F=0.32$ for ABC).
- Catch limits set using target fishing mortality rates below the ABC proxy, and by regulation must be between $F=0.15$ and $F=0.25$.

